



NUST CHIP DESIGN CENTRE

RISC-V Assembly

EXERCISE # 01 **RISCV Assembly Tasks**

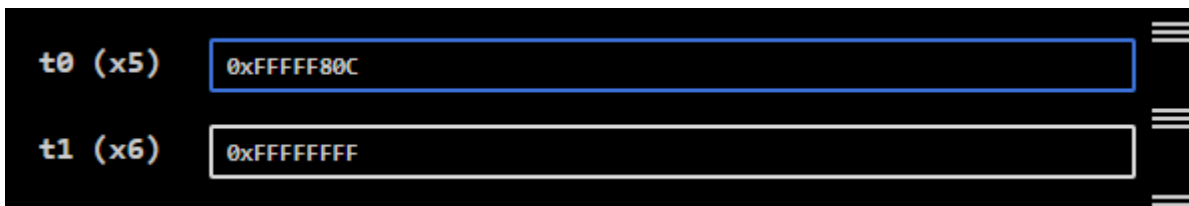
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<u>Instructor</u>	<i><u>Sir Imran & Sir Umer</u></i>
<u>Date</u>	<i><u>13th Aug 2025</u></i>

1. In-Lab Tasks: (Write your lab task & screenshots here)

i. Task 1:

```
.data
## define constant n = 0x7FF (fits in 12-bit imm)
.equ n, 0x7F3
.equ bit, 4
.equ start_lsb_bit, 4
.equ X5, -6
.equ X6, -5

.text
# QUESTION # 01
addi x5, x0, n
addi x6, x0, -1
sub x5, x6, x5
```



ii. Task 2:

```
.data
## define constant n = 0x7FF (fits in 12-bit imm)
.equ n, 0x7F3
.equ bit, 4
.equ start_lsb_bit, 4
.equ X5, -6
.equ X6, -5

.text
# QUESTION # 02
```

RISC-V Assembly

```
addi x5, x0, 0x123
slli x5, x5, 12
addi x5, x5, 0x456
slli x5, x5, 8
addi x5, x5, 0x78
```



iii. Task 3:

```
.data
# # define constant n = 0x7FF (fits in 12-bit imm)
.equ n, 0x7F3
.equ bit, 4
.equ start_lsb_bit, 4
.equ X5, -6
.equ X6, -5

.text
# QUESTION # 03
addi x5, x0, n
slli x10, x5, 31
srli x10, x10, 31
```



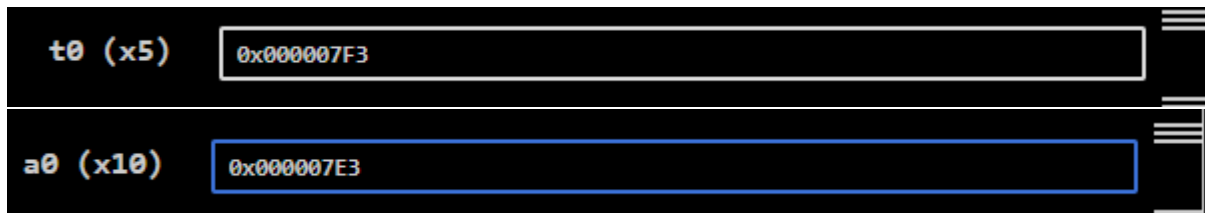
iv. Task 4:

```
.data
# # define constant n = 0x7FF (fits in 12-bit imm)
.equ n, 0x7F3
```

RISC-V Assembly

```
.equ bit, 4
.equ start_lsb_bit, 4
.equ X5, -6
.equ X6, -5
```

```
.text
# QUESTION # 04
addi x5, x0, n
addi x10, x0, 1
slli x10, x10, bit
xor x10, x5, x10
```



v. Task 5:

```
.data
# # define constant n = 0x7FF (fits in 12-bit imm)
.equ n, 0x7F3
.equ bit, 4
.equ start_lsb_bit, 4
.equ X5, -6
.equ X6, -5
```

```
.text
# QUESTION # 05
addi x2, x0, 31 # x2 stores how many 1s
add x3, x0, x0 # x3 loop iterates to 31
add x4, x0, x0 # x4 stores how many 1s
addi x5, x0, n
```

RISC-V Assembly

add x6, x5, x0

addi x8, x0, 1

loop:

andi x9, x6, 1 # if 1 then 1 otherwise 0

bne x9, x8, else # if not equal jump to else

if:

addi x4, x4, 1 # x4 = x4 + 1

else:

srli x6, x6, 1 # x6 shift right

addi x3, x3, 1 # x4 = x4 + 1

beq x3, x2, end # if x3 == 31 end

jal loop

end:

slli x10, x4, 31

srli x10, x10, 31

sp (x2)	0x0000001F
gp (x3)	0x0000001F
tp (x4)	0x00000009
t0 (x5)	0x000007F3
t1 (x6)	0x00000000
t2 (x7)	0x00000000
s0 (x8)	0x00000001
s1 (x9)	0x00000000
a0 (x10)	0x00000001

vi. Task 6:

.data

define constant n = 0x7FF (fits in 12-bit imm)

.equ n, 0x7F3

.equ bit, 4

.equ start_lsb_bit, 4

.equ X5, -6

.equ X6, -5

.text

QUESTION # 06

addi x5, x0, 0x123

slli x5, x5, 12

addi x5, x5, 0x456

slli x5, x5, 8

addi x5, x5, 0x78

RISC-V Assembly

```
addi x4, x0, 0    # bit # 0, 1, 2, 3
addi x6, x0, 1    # bit # 1, 2, 3, 4
addi x7, x0, 2    # bit # 2, 3, 4, 5
addi x8, x0, 3    # bit # 3, 4, 5, 6
addi x9, x0, 4    # bit # 4, 5, 6, 7
# addi x5, x0, n
addi x3, x0, start_lsb_bit
add x10, x5, x0
```

```
beq x3, x4, b0
beq x3, x6, b1
beq x3, x7, b2
beq x3, x8, b3
beq x3, x9, b4
```

```
# beq x3, x4, b0
b0:
slli x10, x5, 16
srli x10, x10, 16
jal end
```

```
# beq x3, x6, b1
b1:
srli x10, x5, 4
slli x10, x10, 16
srli x10, x10, 16
jal end
```

```
# beq x3, x7, b2
b2:
```

RISC-V Assembly

```
srli x10, x5, 8
slli x10, x10, 16
srli x10, x10, 16
jal end
```

```
# beq x3, x8, b3
b3:
srli x10, x5, 12
slli x10, x10, 16
srli x10, x10, 16
jal end
```

```
# beq x3, x9, b4
b4:
srli x10, x5, 16
jal end
```

```
end:
```


sp (x2)	0x7FFFFFFDC
gp (x3)	0x00000004
tp (x4)	0x00000000
t0 (x5)	0x12345678
t1 (x6)	0x00000001
t2 (x7)	0x00000002
s0 (x8)	0x00000003
s1 (x9)	0x00000004
a0 (x10)	0x00001234

vii. Task 7:

.data

define constant n = 0x7FF (fits in 12-bit imm)

.equ n, 0x7F3

.equ bit, 4

.equ start_lsb_bit, 4

.equ X5, -6

.equ X6, -5

.text

QUESTION # 07

addi x5, x0, 0xff

slli x5, x5, 12

addi x5, x5, 0x456

slli x5, x5, 12

addi x5, x5, 0x789

RISC-V Assembly

```
addi x4, x0, 0    # bit # 0, 1, 2, 3
addi x6, x0, 1    # bit # 1, 2, 3, 4
addi x7, x0, 2    # bit # 2, 3, 4, 5
addi x8, x0, 3    # bit # 3, 4, 5, 6
addi x9, x0, 4    # bit # 4, 5, 6, 7
# addi x5, x0, n
addi x3, x0, start_lsb_bit
add x10, x5, x0
```

```
beq x3, x4, b0
beq x3, x6, b1
beq x3, x7, b2
beq x3, x8, b3
beq x3, x9, b4
```

```
# beq x3, x4, b0
b0:
slli x10, x5, 16
srai x10, x10, 16
jal end
```

```
# beq x3, x6, b1
b1:
srli x10, x5, 4
slli x10, x10, 16
srai x10, x10, 16
jal end
```

```
# beq x3, x7, b2
b2:
```

RISC-V Assembly

```
srli x10, x5, 8
slli x10, x10, 16
srai x10, x10, 16
jal end
```

```
# beq x3, x8, b3
b3:
srli x10, x5, 12
slli x10, x10, 16
srai x10, x10, 16
jal end
```

```
# beq x3, x9, b4
b4:
srai x10, x5, 16
jal end
```

```
end:
```

sp (x2)	0x7FFFFFFDC
gp (x3)	0x00000004
tp (x4)	0x00000000
t0 (x5)	0xFF456789
t1 (x6)	0x00000001
t2 (x7)	0x00000002
s0 (x8)	0x00000003
s1 (x9)	0x00000004
a0 (x10)	0xFFFFFFFF45

viii. Task 8:

.data

define constant n = 0x7FF (fits in 12-bit imm)

.equ n, 0x7F3

.equ bit, 4

.equ start_lsb_bit, 4

.equ X5, -6

.equ X6, -5

.text

QUESTION # 08

add x10, x0, x0 # First clear the x10

addi x5, x0, X5

addi x6, x0, X6

blt x6, x5, run

jal end

RISC-V Assembly

run:

addi x10, x0, 1

end:

